

STATEMENT OF NEED
460 CES
Restricted Area Cooling Water Repair Materials
Current as of 17 March 2026

1. SCOPE:

460 CES requires a large-scale material purchase to repair and fortify the cooling water within the Mission Delta 4 (MD4) Restricted Area (RA). Necessary materials include carbon steel pipe repair clamps for leak repair, a pipe freezing kit for temporary isolation for pipes up to an 10" diameter, and carbon fiber cloth for leak hardening.

2. REQUIREMENT:

Product	Quantity
Elbow Repair Package	1
Pipe Freezing Kit	1
Carbon Fiber Cloth Roll, Epoxy and High-Modulus Filler Package	1

3. SPECIFICATIONS:

Elbow Repair Package

- Shall include two (2) 10" pipe repair clamps.
- Shall include two (2) 8" pipe repair clamps.
- Shall include two (2) 6" pipe repair clamps.
- Shall include two (2) 4" pipe repair clamps.
- Shall include two (2) 3" pipe repair clamps.
- Shall include two (2) 10" 90-degree elbow clamps.
- Shall include two (2) 8" 90-degree elbow clamps.
- Shall include two (2) 6" 90-degree elbow clamps.
- Shall include two (2) 4" 90-degree elbow clamps.
- Shall include two (2) 3" 90-degree elbow clamps.
- Shall include two (2) 10" 45-degree elbow clamps.
- Shall include two (2) 8" 45-degree elbow clamps.
- Shall include two (2) 6" 45-degree elbow clamps.
- Shall include two (2) 4" 45-degree elbow clamps.
- Shall include two (2) 3" 45-degree elbow clamps.
- Clamps shall be capable of withstanding up to 60 PSI water pressure.
- Clamps shall be internally 3-Layer Polyethylene (3LPE) coated.
- Clamps shall be carbon steel material.
- All clamps must have either an internal epoxy coating as called out or come with a neoprene sheet.

- Material shall be compliant with ASME PCC-2, ISO 24817, and ASME B13 for sealing high pressure leaks.

Pipe Freezing Kit

- Shall utilize liquid carbon dioxide (CO₂) to freeze stationary water.
- Shall create in-line ice plug within piping system for pipe section isolation.
- Shall form dry ice at a temperature of at minimum -109°F.
- Shall include all necessary jackets to support freezing pipes 3/8" to 10" in diameter.
- Shall include high pressure hoses no less than 16' in length.
- Shall include all necessary CO₂ cylinder valve adapters.
- Shall include all necessary timing boards and pads.
- Material shall be compliant with ASME PCC-2, ISO 24817, and ASME B13 for sealing high pressure leaks.

Carbon Fiber Cloth Roll, Epoxy and High-Modulus Filler Package

- Shall be field-saturated system made of carbon fiber cloth.
- Shall come with and be applied with two-part epoxy.
- Shall be a high-modulus filler material, comparable to industry standard for sealing high-pressure leaks.
- Shall seal steel pipe leaks in both hoop and axial directions.
- Shall operate effectively at temperatures up to 200°F.
- Shall be effective on cooling water lines.
- Shall have ply thickness of at least 0.0105".
- Material shall be compliant with ASME PCC-2, ISO 24817, and ASME B13 for sealing high pressure leaks.

4. DELIVERY:

Must be delivered to Buckley Space Force Base. The contractor shall deliver the above requirements no later than 3 days after receipt of order.

Items, when delivered, will be accepted and inspected by the government officials responsible for receiving the products and performing any required inspections.